

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412999
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Wang; Chao

Storage battery bolt terminal

Abstract

A storage battery bolt terminal, related to the technical field of automobile electric appliance connection including a terminal (100), which includes a storage battery post connecting unit (101). Compressible units (102) are provided on two sides of an opening end of the storage battery post connecting unit (101). A cable connecting unit (104) is provided in another opening end of the storage battery post connecting unit (101). The storage battery bolt terminal further includes a gland (200) with a pressing unit (201), and one side of the gland (200) is provided with a rotating unit (202), wherein the compressible unit (102) is rotatably connected to the rotating unit (202), and the rotating unit (202) is provided with a locking unit (203). The storage battery bolt terminal is convenient to operate and high in working efficiency, tool-free disassembly and assembly are achieved on the connection of the storage battery post, a bolt tightening tool is omitted, the application is flexible, and the mounting cost is reduced.

Inventors:	Wang; Chao (Changchun, CN)
Applicant:	Changchun Jetty Automotive Technology Co., Ltd. (Changchun, CN)
Family ID:	74729756
Assignee:	CHANGCHUN JETTY AUTOMOTIVE TECHNOLOGY CO., LTD. (Changchun, CN)
Appl. No.:	18/012390
Filed (or PCT Filed):	June 25, 2021
PCT No.:	PCT/CN2021/102279
PCT Pub. No.:	WO2022/022184
PCT Pub. Date:	February 03, 2022

Prior Publication Data

Foreign Application Priority Data

CN	202021561663.8	Jul. 31, 2020
----	----------------	---------------

Publication Classification

Int. Cl.: H01R11/28 (20060101)

U.S. Cl.:

CPC H01R11/282 (20130101); H01R11/284 (20130101)

Field of Classification Search

CPC: H01R (11/282); H01R (11/284); H01R (13/447); H01R (4/28); H01R (4/70); H01M (50/50); H01M (50/567); Y02E (60/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3568139	12/1970	Delzer	N/A	N/A
7344421	12/2007	Spencer	439/286	H01R 11/287
9559444	12/2016	Gisoldi	N/A	N/A
2004/0161980	12/2003	Cret	N/A	N/A
2009/0124118	12/2008	Magno, Jr.	439/772	H01R 11/282
2009/0253312	12/2008	Detter et al.	N/A	N/A
2013/0316599	12/2012	Goldschmidt	439/782	H01R 11/282
2023/0268677	12/2022	Wang	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2343683	12/1998	CN	N/A
202121006	12/2011	CN	N/A
103001066	12/2012	CN	N/A
203839536	12/2013	CN	N/A
203967374	12/2013	CN	N/A
208433537	12/2018	CN	N/A
208539134	12/2018	CN	N/A
210027289	12/2019	CN	N/A
212626084	12/2020	CN	N/A
102005049957	12/2006	DE	N/A
0466528	12/1991	EP	N/A
4191795	12/2022	EP	N/A
2600830	12/1986	FR	N/A
2601515	12/1987	FR	N/A
S6263625	12/1986	JP	N/A

S62124859	12/1986	JP	N/A
H10223200	12/1997	JP	N/A
2017059338	12/2016	JP	N/A
2018063781	12/2017	JP	N/A
200443839	12/2008	KR	N/A
20090006548	12/2008	KR	N/A
101984684	12/2018	KR	N/A
2022022184	12/2021	WO	N/A

OTHER PUBLICATIONS

International Search Report from corresponding International Application No.

PCT/CN2021/102279, dated Sep. 27, 2021, 6 pages with translation. cited by applicant

Extended European Search Report from corresponding European Application No. 21851326.5 mailed on Feb. 7, 2024, 10 pages. cited by applicant

Office action from corresponding Japanese Application No. 2023-506007 mailed on Feb. 21, 2024, 12 pages with translation. cited by applicant

Brazilian Office Communication dated Apr. 11, 2025 for Application Serial No. BR112022026960-9 (8 pages). cited by applicant

Korean Office Communication dated Apr. 15, 2025 for Application Serial No. 10-2022-7043501 (16 pages). cited by applicant

European Search report dated Jan. 26, 2024 for Publication No. EP4191795A4 (2 pages). cited by applicant

Canadian Office Action dated Jul. 8, 2025 for Application Serial No. 3,182,209 (6 pages). cited by applicant

Primary Examiner: Duverne; Jean F

Attorney, Agent or Firm: Tucker Ellis, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present disclosure is a National State of International Application No.

PCT/CN2021/102279, filed Jun. 25, 2021, which claims priority to Chinese Patent Application No. 202021561663.8, filed on Jul. 31, 2020, and entitled “storage battery bolt terminal”, which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to a technical field of automobile electric appliance connection, and particularly to a quick replaceable storage battery bolt connection terminal.

BACKGROUND

(3) The automobile storage battery bolt terminal is an indispensable component for connecting the automotive wiring harness to the storage battery and is necessary for every automobile. The important function of the automobile storage battery bolt terminal is to achieve the connection between the wiring harness of the whole vehicle and the storage battery, so that the electric functions of the whole vehicle can be normal. It is necessary to mount the storage battery bolt reliably, safely and conveniently. The existing storage battery bolt terminal is firstly disposed to sleeve a storage battery post and then is tightened and secured with a bolt, or when the storage battery bolt terminal needs to be disassembled to cut off the power supply in case of maintenance

or transportation of the whole vehicle, the operation must be completed with a tool, which is very inconvenient and there is a safety risk in the hot-line works during maintenance.

SUMMARY

(4) Aiming at the defects in the prior art, the problem to be solved herein is to provide a storage battery bolt terminal which is convenient and achieves tool-free disassembly and assembly, so as to overcome the technical defect caused by the fact that the existing storage battery bolt terminal needs to be operated by a bolt tightening tool during disassembly and assembly.

(5) The present disclosure adopts the following technical solution to solve the technical problem: there is provided a storage battery bolt terminal, including: a terminal including a storage battery post connecting unit, compressible units being provided on two sides of an opening end of the storage battery post connecting unit and a cable connecting unit being provided in another opening end of the storage battery post connecting unit; and a gland including a pressing unit and two rotating units provided on the pressing unit (201), wherein the compressible unit is rotatably connected to the rotating unit, and the rotating unit is provided with a locking unit. The storage battery post connecting unit is connected to a storage battery post, with a conical shape matched with that of the storage battery post, and a size matched with that of the storage battery post, and may be matched with and connected to the anode and the cathode of the storage battery as required.

(6) A further solution of the present disclosure is that the gland is externally provided with a protective cover adapted to the pressing unit, and the protective cover is fixedly connected to the gland by clamping. The protective cover includes a main unit and a clamping unit, and the main unit of the protective cover is of rectangular shape, or may be of any circular or polygonal shape as long as it is matched with the pressing unit. The function of the clamping unit is to clamp the gland without loosening or falling off. There may be one clamping unit at the top and the bottom, respectively, or the number of the clamping units may also be three, four, five, etc.

(7) A further solution of the present disclosure is that when the gland is rotated to the above of the terminal, the locking unit locks the terminal with a storage battery post by compressing the compressible unit. During disassembly of the storage battery bolt terminal, the locking unit can be released from the compressible unit once the protective cover is raised by hand, thus realizing the disassembly between the storage battery bolt terminal and the storage battery post, which is very convenient and quick. The storage battery post connecting unit is connected to the storage battery post and matched with the shape thereof.

(8) A further solution of the present disclosure is that a stop unit is provided outside the compressible unit, and when the gland is rotated, the rotating unit abuts against the stop unit to form a limiting fit between the stop unit and the rotating unit. When the protective cover and the gland are rotated to a certain degree, the rotation of the gland stops. At the same time, the storage battery post connecting unit is connected to the storage battery post.

(9) A further solution of the present disclosure is that the protective cover is made of a flame retardant hard material. Different types of materials, such as PBT-GF30, PA6, PA66, PC/ABS, etc., are selected for the protective cover depending on strengths, temperature-resistant grades, and material characteristics, and any material should meet the requirements of flame retardance and strength.

(10) A further solution of the present disclosure is that the terminal is integrally formed.

(11) A further solution of the present disclosure is that the gland is integrally formed.

(12) The terminal and the gland are both integrally formed, ensuring that the storage battery bolt terminal is firmer. Different types of materials, such as red copper, brass, lead brass, etc., may be selected for the terminal depending on the machining process, electrical conductivity and material characteristics. The surface of the terminal is plated with tin, silver, nickel, etc. depending on the environmental requirements, thus meeting the requirements of electrical conductivity and corrosion resistance. The gland may be made of red copper, brass, lead brass, carbon steel, low-carbon steel, stainless steel, etc. depending on the machining process. In order to meet the requirement of

corrosion resistance, the surface of the copper-containing material may be plated with tin, silver and nickel depending on environmental requirements, and the surface of the carbon steel may be plated with zinc, zinc-nickel alloy, etc.

(13) A further solution of the present disclosure is that the pressing unit is of any circular or polygonal shape.

(14) A further solution of the present disclosure is that the pressing unit is of rectangular shape.

(15) The pressing unit is clamped and matched with the protective cover, and the pressing unit may be pressed and raised in a process of pressing and raising the protective cover.

(16) A further solution of the present disclosure is that the cable connecting unit includes at least one wiring hole, and a direction of the wiring hole may be 0° to 180° according to an outgoing line direction.

(17) A further solution of the present disclosure is that the cable connecting unit is connected to the cable by crimping, welding or screwing.

(18) The cable connecting unit is connected to the cable, and has a shape that matches the profile of the wire. An alternative profile of the cable connecting unit is a circular or any polygonal shape. According to the connection mode and requirement of the cable, crimping, welding or screwing may be adopted. Depending on the number of the cables to be connected, the cable connecting unit may have one, two or three wiring holes, etc. Depending on the outgoing line direction, the opening direction of the wiring holes may be 0° to 180° .

(19) A further solution of the present disclosure is that the compressible unit is hinged to the rotating unit by a rotating shaft. The rotating shaft connects the terminal and the gland to play roles of connection and rotation. The rotating shaft includes a central shaft cap unit, a central shaft unit and a plug unit, wherein the central shaft cap unit is of circular shape to prevent the rotating shaft from falling off, and an alternative profile is of circular or polygonal shape such as ellipse shape, square shape, etc. The central shaft unit is of cylindrical shape to play roles of connection and rotation. The plug unit is of concave circular shape to prevent the gland and the terminal from falling off.

(20) The present disclosure has the following advantageous effects. 1. By using the storage battery bolt terminal, the disconnection and connection of the storage battery bolt terminal can be realized by simple manual operations, and assembly or disassembly of the storage battery bolt terminal and the storage battery post can be achieved by pressing or raising the protective cover, so that the bolt tightening tool is omitted, the maintenance convenience is greatly improved, the structure is simple and practical, the manufacturing process is simple, and the mounting cost is reduced, thus having outstanding practical values. 2. In the storage battery bolt terminal, the protective cover is connected to the gland by clamping, and the protection cover has a flame retardant and insulating performance. During maintenance, the storage battery bolt can be disconnected by simple manual operation, and there is no safety risk when pressing and raising the protection cover by hand, so the safety is greatly improved. 3. The cable connecting unit is connected to the cable by welding, crimping and screwing. A connection in any direction of 0° to 180° can be realized depending on the outgoing line direction, thus achieving different connection modes and more flexible applications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be further described below with reference to the drawings and the embodiments. In the drawings,

(2) FIG. 1 is a three-dimensional structural diagram of a storage battery bolt terminal according to the present disclosure;

- (3) FIG. 1-1 is a diagram of a semi-closed state of a gland of a storage battery bolt terminal according to the present disclosure;
- (4) FIG. 1-2 is a diagram of a closed state of a gland of a storage battery bolt terminal according to the present disclosure;
- (5) FIGS. 2, 2-1 and 2-2 are three-dimensional structural diagrams of a gland according to the present disclosure;
- (6) FIGS. 3 and 3-1 to 3-6 are three-dimensional structural diagrams of a terminal according to the present disclosure;
- (7) FIGS. 4 and 4-1 are structural diagrams of a protective cover according to the present disclosure;
- (8) FIG. 5 is a three-dimensional structural diagram of a rotating shaft according to the present disclosure;
- (9) FIGS. 6-1 and 6-2 are screwing structural diagrams of a cable connecting unit according to the present disclosure;
- (10) FIGS. 7-1, 7-2 and 7-3 are diagrams of an open state, a semi-compressed state, and a locked state of a terminal according to the present disclosure, respectively.
- (11) FIG. 8-1 illustrates, from left to right, outgoing line directions of 0°, 45°, 90°, 135° and 180° on a YZ plane; and
- (12) FIG. 8-2 illustrates, from left to right, outgoing line directions of 0°, 45°, 90°, 135° and 180° on an XY plane.

REFERENCE NUMERALS

- (13) **100**: terminal; **200**: gland; **300**: protective cover; **400**: rotating shaft; **101**: storage battery post connecting unit; **102**: compressible unit; **103**: stop unit; **104**: cable connecting unit; **201**: pressing unit; **202**: rotating unit **203**: locking unit; **301**: main unit; **302**: clamping unit; **401**: shaft cap unit; **402**: central shaft unit; **403**: plug unit.

DETAILED DESCRIPTION

- (14) Exemplary embodiments of the present disclosure will now be described in detail with reference to the drawings.

Embodiments of the First Aspect

- (15) As illustrated in FIGS. 1 to 5, a storage battery bolt terminal according to an embodiment of the present disclosure includes a terminal **100** including a storage battery post connecting unit **101**, compressible units **102** being provided on two sides of an opening end of the storage battery bolt connecting unit **101**, and a cable connecting unit **104** being provided in another opening end of the storage battery post connecting unit **101**. The storage battery bolt terminal further includes a gland **200**, which includes a pressing unit **201** and two rotating units **202** provided on the pressing unit **201**.

- (16) The compressible unit **102** is rotatably connected to the rotating unit **202**, and the rotating unit **202** is provided with a locking unit **203**. The storage battery post connecting unit **101** is connected to a storage battery post, with a conical shape matched with that of the storage battery post, and a size matched with and greater than that of the storage battery post.

- (17) Further, in order to protect the safety of workers and prevent the workers from directly touching the gland, the gland **200** is externally provided with a protective cover **300** adapted to the pressing unit **201**.

- (18) In some implementations, the protective cover **300** is fixedly connected to the gland **200** by clamping. As illustrated in FIG. 4, the protective cover includes a main unit **301** and a clamping unit **302**, and the main unit of the protective cover **300** is of rectangular shape. As illustrated in FIGS. 2-1, 2-2 and 4-1, the gland **200** is of circular or triangular shape, and the protective cover **300** is of circular shape. It can be understood that the gland is of any circular or polygonal shape, and the protective cover **300** may be of any circular or polygonal shape as long as it is matched with the pressing unit **201**. The protective cover **300** should cover the gland **200** to ensure that the

gland is not touched by human hands when the storage battery bolt terminal is in operation. The function of the clamping unit **302** is to clamp the gland **200** without loosening or falling off. There may be one clamping unit **302** at the top and the bottom, respectively, or the number of the clamping units **302** may also be three, four, five, etc.

(19) In other implementations, the protective cover **300** and the gland **200** may be connected in other ways, which is not limited herein.

(20) According to the embodiment, the disconnection and connection of the storage battery bolt terminal can be realized by simple manual operations, and assembly or disassembly of the storage battery bolt terminal and the storage battery post can be achieved by pressing or raising the protective cover, so that the bolt tightening tool is omitted, the maintenance convenience is greatly improved, the structure is simple and practical, the manufacturing process is simple, and the mounting cost is reduced, thus having outstanding practical values.

(21) As illustrated in FIGS. **1-1** and **1-2**, the storage battery post connecting unit is a sleeve hole structure (e.g., a cylindrical hole), and the sleeve hole of the storage battery post connecting unit is disposed to sleeve the storage battery post, wherein the storage battery post is a conical cylinder.

(22) As illustrated in FIG. **2**, the locking unit **203** is provided on one of the rotating units **202**. The locking unit **203** is a strip-shaped structure protruding from the rotating unit **202**, and a side of the locking unit **203** close to a rotating end of the rotating unit is an inclined surface.

(23) In some implementations, in order to improve the efficiency of the locking unit **203** compressing the compressible unit **102**, a position where the locking unit **203** contacts the compressible unit **102** is a transition corner, by which the opening between the compressible units **102** is gradually narrowed by compressing. The transition corner mentioned herein refers to a rounded arc corner.

(24) When the protective cover **300** is rotatably pressed, the gland **200** is rotated toward the terminal **100**, and when the gland **200** is rotated to the above of the terminal **100**, the locking unit **203** compresses the compressible unit **102** to shrink the opening between the compressible units **102**, thus locking the terminal **100** with the storage battery post, and realizing the connection and contact between the storage battery bolt terminal and the storage battery post.

(25) Specifically, as illustrated in FIGS. **7-1** to **7-3**, the specific process of connecting the storage battery bolt terminal to the storage battery post includes: firstly, the gland **200** is in an open state, and the locking unit **203** does not compress the compressible unit **102**, as illustrated in FIG. **7-1**; in this state, the protective cover **300** is rotatably pressed to drive the gland **200** to be rotated towards the terminal **100**, the locking unit **203** compresses the compressible unit **102**, and the opening between the compressible units **102** shrink; as illustrated in FIG. **7-2**, the protective cover **300** is continued to be rotatably pressed to drive the gland **200** to be rotated towards the terminal **100**, and the gland **200** drives the locking unit **203** to compress the compressible unit **102** until the protective cover **300** cannot move, thus locking the terminal **100** and the storage battery post.

(26) During disassembly of the storage battery bolt terminal, since the protective cover **300** is clamped with the gland **200**, the locking unit **203** can be released from the compressible unit once the protective cover **300** is raised by hand, thus realizing the disassembly between the storage battery bolt terminal and the storage battery post, which is very convenient and quick. The storage battery post connecting unit **101** is connected to the storage battery post and matched with the shape thereof.

(27) In some implementations, a stop unit **103** is provided outside the compressible unit **102**, and the stop unit **103** and the locking unit **203** are disposed on opposite sides. During mounting of the storage battery post, the protective cover **300** is pressed to drive the gland **200** to be rotated, so that the rotating unit **202** abuts against the stop unit **103**, thus forming a limiting fit between the stop unit **103** and the rotating unit **202**. When the protective cover and the gland are rotated to a certain degree, the rotation of the gland **200** stops. At the same time, the storage battery post connecting unit **101** is connected to the storage battery post.

(28) The embodiment can prevent the excessive rotation of the gland, and further avoid the phenomenon that the gland contacts the storage battery post to conduct electricity.

(29) As illustrated in FIG. 4, the protective cover **300** is made of a flame retardant hard material. Different types of materials, such as PBT-GF30, PA6, PA66, PC/ABS, etc., are selected for the protective cover **300** depending on strengths, temperature-resistant grades, and material characteristics, and any material should meet the requirements of flame retardance and strength.

(30) The terminal **100** and the gland **200** are both integrally formed, ensuring that the storage battery bolt terminal is firmer. Different types of materials, such as red copper, brass, lead brass, etc., may be selected for the terminal **100** depending on the machining process, electrical conductivity and material characteristics. The surface of the terminal **100** is plated with tin, silver, nickel, etc. depending on the environmental requirements, thus meeting the requirements of electrical conductivity and corrosion resistance. The gland **200** may be made of red copper, brass, lead brass, carbon steel, low-carbon steel, stainless steel, etc. depending on the machining process. In order to meet the requirement of corrosion resistance, the surface of the copper-containing material may be plated with tin, silver and nickel depending on environmental requirements, and the surface of the carbon steel may be plated with zinc, zinc-nickel alloy, etc.

(31) As illustrated in FIGS. 2, 2-1 and 2-2, the structure of the pressing unit **201** may be of any circular or polygonal shape, and exemplarily, the structure of the pressing unit **201** is of rectangular shape. The pressing unit **201** is clamped and matched with the protective cover **300**, and the pressing unit **201** may be pressed and raised in the process of pressing and raising the protective cover **300**.

(32) As illustrated in FIGS. 3 and 3-1 to 3-6, the cable connecting unit **104** includes at least one wiring hole, and crimping or welding may be adopted according to the connection mode of the cable. According to the outgoing line direction, the direction of the wiring hole of the cable connecting unit **104** may be 0° to 180° . The cable connecting unit **104** is connected to the cable, and has a circular profile that matches the shape of the wire. The alternative profile is a circular or any polygonal shape. According to the connection mode of the cable, crimping or welding may be adopted. Depending on the number of the cables connected, the cable connecting unit **104** may have one, two or three wiring holes, etc. Depending on the outgoing line direction, the opening direction of the wiring holes may be 0° to 180° . FIGS. 3-1 to 3-3 are diagrams of terminal turning, and the opening direction of the wiring hole may be 0° or 180° . As illustrated in FIGS. 3-4 to 3-6, the opening direction of the wiring hole is 90° , and may be adjusted to any direction of 0° to 180° depending on the outgoing line direction.

(33) Specifically, the adjustment range of the direction of the wiring hole is 0° to 180° on a vertical plane where the cable connecting unit is located (as illustrated in FIG. 8-1) and/or 0° to 180° on a horizontal plane where the cable connecting unit is located (as illustrated in FIG. 8-2).

(34) Specifically, the vertical plane where the cable connecting unit is located refers to a plane formed by adjustable paths of the cable connecting unit in a vertical direction (abbreviated as a YZ plane), and as illustrated in FIG. 8-1, which illustrates, from left to right, the outgoing line directions of 0° , 45° , 90° , 135° and 180° on the YZ plane.

(35) The horizontal plane where the cable unit is located refers to a plane formed by adjustable paths of the cable in a horizontal direction (abbreviated as an XY plane), and as illustrated in FIG. 8-2, which illustrates, from left to right, the outgoing line directions of 0° , 45° , 90° , 135° and 180° on the XY plane.

(36) In some implementations, as illustrated in FIG. 5, the compressible units **102** are hinged to the rotating units **202** by a rotating shaft **400**, and the rotating shaft **400** connects the terminal **100** and the gland **200** to play roles of connection and rotation. The rotating shaft **400** includes a central shaft cap unit **401**, a central shaft unit **402** and a plug unit **403**, wherein the central shaft cap unit **401** is of circular shape to prevent the rotating shaft from falling off, and the alternative profile is of circular or polygonal shape such as ellipse shape, square shape, etc. The central shaft unit **402** is of

cylindrical shape to play roles of connection and rotation. The plug unit **403** is of concave circular shape and is detachably provided on the central shaft unit **402** to prevent the gland **200** and the terminal **100** from falling off.

(37) During implementation, firstly, the plug unit **403** is separated from the central shaft unit **402**, and the central shaft unit **402** is enabled to pass through mounting holes of the rotating unit **202** and the compressible unit **102**. The plug unit **403** is interference-fitted into the mounting hole of the central shaft unit **402**, or screwed into a threaded hole of the central shaft unit **402**.

(38) The embodiments have a characteristic of simple structure, which facilitates the mounting of the compressible unit and the rotating unit.

Embodiments of the Second Aspect

(39) As illustrated in FIGS. **1** to **6**, the storage battery bolt terminal according to an embodiment includes a terminal **100**. The terminal **100** includes a storage battery post connecting unit **101**, compressible units **102** provided on two sides of an opening end of the storage battery bolt connecting unit **101**, and a cable connecting unit **104** in another opening end of the storage battery post connecting unit **101**. The storage battery bolt terminal further includes a gland **200**, which includes a pressing unit **201**, and two rotating units **202** provided on a side of the gland **200** and on the pressing unit **201**.

(40) The compressible unit **102** is rotatably connected to the rotating unit **202**, and the rotating unit **202** is provided with a locking unit **203**. The storage battery post connecting unit **101** is connected to a storage battery post, with a conical shape matched with that of the storage battery post, and a size matched with and greater than that of the storage battery post. The gland **200** is externally provided with a protective cover **300** adapted to the pressing unit **201**, and the protective cover **300** is fixedly connected to the gland **200** by clamping. The protective cover includes a main unit **301** and a clamping unit **302**, and the main unit of the protective cover **300** is of rectangular shape. As illustrated in FIGS. **2-1**, **2-2** and **4-1**, the gland **200** is of circular or triangular shape, and the protective cover **300** is of circular shape. It can be understood that the gland is of any round or polygonal shape, and the protective cap **300** may be of any circular or polygonal shape as long as it is matched with the pressing unit **201**. The function of the clamping unit **302** is to clamp the gland **200** without loosening or falling off. There may be one clamping unit **302** at the top and the bottom, respectively, or the number of the clamping units **302** may also be three, four, five, etc.

(41) After the storage battery bolt terminal is mounted to sleeve the storage battery post, when the protective cover **300** is rotatably pressed, the gland **200** is rotated to the above of the terminal **100**, the locking unit **203** compresses the compressible unit **102** to lock the terminal **100** with the storage battery post. During disassembly of the storage battery bolt terminal, since the protective cover **300** is clamped with the gland **200**, the locking unit **203** can be released from the compressible unit once the protective cover **300** is raised by hand, thus realizing the disassembly between the storage battery bolt terminal and the storage battery post, which is very convenient and quick. The storage battery post connecting unit **101** is connected to the storage battery post and matched with the shape thereof. A stop unit **103** is provided outside the compressible unit **102**, and the stop unit **103** and the locking unit **203** are disposed on opposite sides. During mounting of the storage battery post, the protective cover **300** is pressed to drive the gland **200** to be rotated, so that the rotating unit **202** abuts against the stop unit **103**, thus forming a limiting fit between the stop unit **103** and the rotating unit **202**. When the protective cover and the gland are rotated to a certain degree, the rotation of the gland **200** stops. At the same time, the storage battery post connecting unit **101** is connected to the storage battery post.

(42) As illustrated in FIG. **4**, the protective cover **300** is made of a flame retardant hard material. Different types of materials, such as PBT-GF30, PA6, PA66, PC/ABS, etc., are selected for the protective cover **300** depending on strengths, temperature-resistant grades, and material characteristics, and any material should meet the requirements of flame retardance and strength.

(43) The terminal **100** and the gland **200** are both integrally formed, ensuring that the storage

battery bolt terminal is firmer. Different types of materials, such as red copper, brass, lead brass, etc., may be selected for the terminal **100** depending on the machining process, electrical conductivity and material characteristics. The surface of the terminal **100** is plated with tin, silver, nickel, etc. depending on the environmental requirements, thus meeting the requirements of electrical conductivity and corrosion resistance. The gland **200** may be made of red copper, brass, lead brass, carbon steel, low-carbon steel, stainless steel, etc. depending on the machining process. In order to meet the requirement of corrosion resistance, the surface of the copper-containing material may be plated with tin, silver and nickel depending on environmental requirements, and the surface of the carbon steel may be plated with zinc, zinc-nickel alloy, etc.

(44) As illustrated in FIGS. **2**, **2-1** and **2-2**, the pressing unit **201** may be of any circular or polygonal shape, and exemplarily, the pressing unit **201** is of rectangular shape. The pressing unit **201** is clamped and matched with the protective cover **300**, and the pressing unit **201** may be pressed and raised in the process of pressing and raising the protective cover **300**.

(45) As illustrated in FIGS. **6-1** and **6-2**, the cable connecting unit **104** and the cable are connected by screwing. As illustrated in FIG. **5**, the compressible unit **102** is hinged to the rotating unit **202** by a rotating shaft **400**, which connects the terminal **100** and the gland **200** to play roles of connection and rotation. The rotating shaft **400** includes a central shaft cap unit **401**, a central shaft unit **402** and a plug unit **403**, wherein the central shaft cap unit **401** is of circular shape to prevent the rotating shaft from falling off, and the alternative profile is of circular or polygonal shape such as ellipse shape, square shape, etc. The central shaft unit **402** is of cylindrical shape to play roles of connection and rotation. The plug unit **403** is of concave circular shape to prevent the gland **200** and the terminal **100** from falling off.

(46) It should be understood that the above embodiments are only used to illustrate the technical solutions of the present disclosure, rather than making limitations thereto. For those skilled in the art, the technical solutions in the above embodiments can be modified, or some technical features thereof can be equivalently replaced, and all these modifications and replacements should fall within the protection scope of the claims of the present disclosure.

Claims

1. A storage battery bolt terminal, comprising: a terminal comprising a storage battery post connecting unit, a first and second compressible units being provided on two sides of an opening end of the storage battery post connecting unit, and a cable connecting unit being provided in another opening end of the storage battery post connecting unit; and a gland comprising a pressing unit and a first and second rotating units provided on the pressing unit, wherein the first and second compressible units are rotatably connected to the first and second rotating units, and the first rotating unit is provided with a locking unit, which is located outside the first compressible unit; wherein the gland is externally provided with a protective cover adapted to the pressing unit, and the protective cover is fixedly connected to the gland; when the gland is rotated toward the terminal, the locking unit locks the terminal with a storage battery post by compressing the compressible unit; and a stop unit is provided outside the second compressible unit and protrudes from an outer side surface of the storage battery post connecting unit towards the second rotating unit, the stop unit and the locking unit are disposed on opposite sides, and when the gland is rotated, the second rotating unit abuts against a side of the stop unit that faces away from the storage battery post connecting unit to form a limiting fit between the stop unit and the second rotating unit, to prevent the gland from being in contact with the storage battery post.
2. The storage battery bolt terminal according to claim 1, wherein a position where the locking unit contacts the compressible unit is a transition corner, by which the opening between the first and second compressible units is gradually narrowed by compressing.
3. The storage battery bolt terminal according to claim 1, wherein the protective cover is made of a

flame retardant hard material.

4. The storage battery bolt terminal according to claim 1, wherein the terminal is integrally formed.
 5. The storage battery bolt terminal according to claim 1, wherein the gland is integrally formed.
 6. The storage battery bolt terminal according to claim 1, wherein the pressing unit is of any circular or polygonal shape.
 7. The storage battery bolt terminal according to claim 6, wherein the pressing unit is of rectangular shape.
 8. The storage battery bolt terminal according to claim 7, wherein the cable connecting unit comprises at least one wiring hole, and a direction of the wiring hole is adjusted according to an outgoing line direction.
 9. The storage battery bolt terminal according to claim 8, wherein an adjustment range of the direction of the wiring hole is 0° to 180° on a horizontal plane (XY) where the cable connecting unit is located, and/or 0° to 180° on a vertical plane (YZ) where the cable connecting unit is located.
 10. The storage battery bolt terminal according to claim 8, wherein the cable connecting unit is connected to a cable by crimping, welding or screwing.
 11. The storage battery bolt terminal according to claim 10, wherein the first and second compressible units are hinged to the first and second rotating units by a rotating shaft.
-